

DYNAMIC DISASTER COMMUNICATION HUB

A MINI PROJECT REPORT

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Certified that this project report titled “**DYNAMIC DISASTER COMMUNICATION HUB**” is the bonafide work of **YUGENDAR.B**, II Year, Department of Electronics Engineering (VLSI Design and Technology), who carried out the project work under my supervision.

Certified further that, to the best of my knowledge, the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion, on this or any other candidate.

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ABSTRACT

Disasters turn communications into a life-safety capability. Floodwater, cyclone winds, earthquake damage, landslides, power loss, road isolation, transport-link failure, and network congestion can remove access to alerts, rescue coordination, health requests, and situational reports. The proposed Dynamic Disaster Communication Hub uses an ESP32 controller, an external regional-frequency-compliant LoRa transceiver, local storage, battery or solar power, and a controlled mesh/store-and-forward protocol. The design supports short text and status records, authenticated identities, priority queues, message expiry, delayed delivery, and optional local access through Wi-Fi or Bluetooth. The system is deliberately layered. Raw LoRa provides the infrastructure-free radio link. A message layer makes records durable and deduplicated. A routing layer selects multi-hop, opportunistic, or store-carry-forward delivery. An AI-assisted edge layer performs bounded tasks such as sensor-health checks, anomaly screening, and message triage. Human operators remain accountable for evacuation, dispatch, and hazardous action. The main contribution is a realistic architecture for graceful degradation, not a promise of universal coverage. The report explains why ordinary LoRaWAN is not itself a peer mesh, how low data rate and congestion constrain the design, and how security, privacy, interoperability, energy, and evaluation should be treated as first-class engineering requirements.

Keywords: disaster management; Internet of Things; artificial intelligence; ESP32; LoRa; delay-tolerant networking; mesh networking; emergency communication; store-and-forward; edge computing.

Report map: Pages 3–5 define the need and requirements. Pages 6–7 explain the communication technologies. Pages 8–14 present the proposed system. Pages 15–17 specify implementation and evaluation. Pages 18–20 discuss implications, conclusion, and references.

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence
IoT – Internet of Things
ESP32 – Espressif 32-bit microcontroller family
LoRa – Long Range radio technology
BLE – Bluetooth Low Energy
Wi-Fi – Wireless Fidelity
RF – Radio Frequency
DTN – Delay-Tolerant Networking
P0 – Priority class 0 / life safety
P1 – Priority class 1 / medical or evacuation
P2 – Priority class 2 / logistics
P3 – Priority class 3 / sensor status
P4 – Priority class 4 / administration
TTL – Time To Live
PDR – Packet Delivery Ratio
RSSI – Received Signal Strength Indicator
SNR – Signal-to-Noise Ratio
LMR – Land Mobile Radio
API – Application Programming Interface
GPS – Global Positioning System

CHAPTER 1

INTRODUCTION

The problem addressed in this report is specific: **How can a local system maintain low-bandwidth, authenticated, prioritized information exchange when cellular and internet infrastructure is unavailable or intermittent?** The answer proposed here is not to recreate a national network. It is to deploy small, inexpensive nodes that can discover neighbors, persist messages, relay them across multiple hops, and deliver them when a path becomes available.

The project objectives are to: (1) define an all-hazards communication failure model; (2) design a low-cost ESP32-plus-LoRa node; (3) specify a decentralized message and routing layer; (4) include practical AI and IoT functions without autonomous high-risk decisions; (5) define security, privacy, interoperability, and power requirements; and (6) create a test plan that measures delivery, latency, energy, coverage, and graceful degradation.

The scope excludes voice and video as primary services because LoRa links are low-rate. It also excludes guaranteed emergency delivery, universal interoperability, autonomous evacuation orders, and claims of fixed radio range. All proposed performance values must be measured within a declared geography, traffic load, node density, and regulatory setting.

1.2 Disaster Context and Failure Analysis

A disaster produces a chain of failures rather than a single outage. Flooding can enter equipment rooms, cut roads, interrupt power, and isolate repair crews. Cyclone winds can damage towers, antennas, fiber, and local distribution. Earthquakes can destroy buildings and create aftershocks, rubble, and overloaded networks. Rainfall-triggered landslides can sever roads and bury cables. Even when radio equipment remains physically intact, batteries and generators may be exhausted, transport links may fail, or a surviving network may be overloaded by simultaneous demand.

Hurricane Helene reporting illustrates the combined nature of the problem. The Congressional Research Service identifies roads, power, towers, and fiber transport links as mutually reinforcing constraints on restoration [3](#). An FCC status report recorded outages attributed to damage, transport failure, and power failure [4](#). FEMA's review of the 2017 hurricane season reported that 95% of Puerto Rico's cell towers were out of service after Hurricane Maria and that outages persisted for months [5](#). USGS notes that landslides and debris flows can affect rivers, roads, and structures, directly impeding rescue access and communications restoration [6](#).

The design implication is that redundancy must include more than duplicate radios. It must include alternate power, elevated relay placement, offline forms, trained personnel, paper contact lists, access routes, and procedures for reconnecting to official systems. The proposed hub is one local layer in that wider continuity plan.

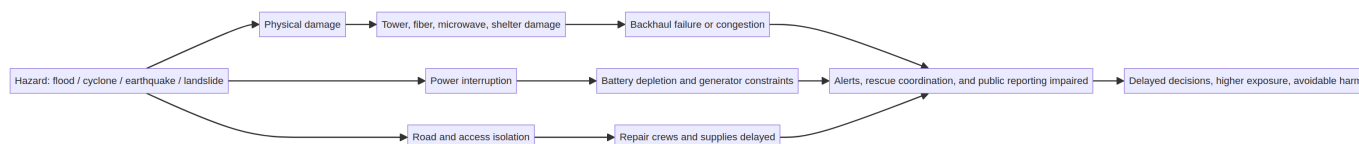


Figure 1. Cascading hazard-to-communication failure model. Rendered from Mermaid source.

Hazard or stressor	Immediate failure	Public-safety consequence	Design response
Flood / cyclone	Power, tower, road, and fiber damage	Alerts and rescue coordination delayed	Elevated relays, battery, offline messages
Earthquake	Physical destruction and overload	Conflicting reports and isolation	Local peer exchange, prioritization
Landslide	Road and backhaul severed	Repair and rescue access blocked	Store-carry-forward, mobile relays
Congestion	Surviving network saturated	Calls and reports do not complete	Compact messages and admission control

1.3 Operational Requirements and Use Cases

The hub must serve several user groups without assuming that each group has the same device, authority, or training. A resident may use a local phone interface or a simple shelter terminal to send a distress message. A responder may carry a relay node and submit structured reports. A health post may report medicine or oxygen needs. A shelter may publish occupancy and water status. An incident commander may receive prioritized summaries when a gateway or carried node returns to the command post.

The minimum information unit should be a compact, signed record rather than an open-ended chat message. It should contain a stable message identifier, sender role, creation time, approximate location or location uncertainty, priority, message type, expiry time, provenance, and integrity protection. A free-text field may be included, but the interface should encourage structured categories such as **life safety**, **medical**, **evacuation**, **shelter**, **logistics**, **sensor event**, and **administrative**.

CHAPTER 2

COMMUNICATION TECHNOLOGIES AND NETWORKING

Accessibility is part of reliability. Interfaces should use plain language, support local languages where feasible, avoid color-only status cues, and provide a manual or paper fallback. The system must not require every affected person to own a smartphone. Community volunteers and trained responders should be able to enter reports on behalf of others while recording the source and uncertainty.

Stakeholder	Primary use case	Information need	Priority
Resident / volunteer	Distress, welfare, missing person	Location, category, short description	Highest
Responder	Situation report	Hazards, access, casualties, resources	High
Health post	Medical request	Patient load, supplies, urgency	High
Shelter	Capacity and utilities	Occupancy, water, power, safety	Medium–high
Utility team	Restoration status	Damage, crew needs, route condition	Medium
Incident command	Coordination	Verified summaries, uncertainty, queues	High

Related technologies and design rationale

The classic ESP32 is a 2.4 GHz Wi-Fi and Bluetooth/Bluetooth Low Energy system-on-chip. It does **not** contain a LoRa or sub-GHz radio, so a real node requires an external LoRa transceiver, suitable antenna, matching RF layout, and regionally compliant frequency and power settings [7](#). ESP32 remains useful because it can host local forms, manage sensors, store records, and control sleep and wake behavior.

The energy model must include the complete board. Espressif’s measurements show substantially higher current during Wi-Fi transmission than a sleeping controller, while deep sleep powers down wireless peripherals and does not maintain Wi-Fi or Bluetooth connections [7](#). A node that needs continuous Wi-Fi access therefore has a different power budget from a node that wakes periodically for LoRa messages.

LoRa offers long-range, low-rate communication. Semtech describes favorable distances that depend strongly on environment and line of sight [9](#). Its data-rate and payload trade-offs mean that the system should carry short text, status, and sensor records rather than voice, images, or interactive video [10](#).

LoRaWAN must be distinguished from raw LoRa. LoRaWAN is a star-of-stars architecture in which end devices communicate with gateways and a network server [11](#). It is not automatically an infrastructure-free peer mesh. An offline deployment may use a locally hosted gateway and server, but a truly decentralized field layer needs raw LoRa or a custom local protocol. Capacity also falls as airtime, node count, and collisions increase [12](#).

Link option	Infrastructure assumption	Best use in this project	Principal limitation
Raw LoRa	Neighbor nodes and local protocol	Offline text and relay	Custom protocol and regulation
LoRaWAN	Gateway and network server	Local managed sensor network	Not peer-to-peer by default
Wi-Fi/BLE	Local device proximity	Phone or terminal access	Higher energy / short range
Cellular	Working operator network	Backhaul when available	Towers, power, congestion
Satellite / LMR	Specialized equipment	Last-resort or responder link	Cost, equipment, access

Mesh and delay-tolerant networking concepts

A mesh is not simply a group of radios. It is a set of forwarding rules that determine which node relays which message, how duplicates are suppressed, how long records remain valid, and what happens when the network is partitioned. In a connected neighborhood, a message can move through multiple hops. In a partition, a node may store the record until a responder, vehicle, or mobile relay creates a contact.

Delay-tolerant networking (DTN) formalizes this idea as message-oriented, store-and-forward communication over intermittent links [13](#). Bundle Protocol version 7 allows contacts to be scheduled, predicted, or opportunistic and separates application identity from changing

underlying addresses 14. That separation is valuable after a disaster because the node that can deliver a message may not be known when the record is created.

The proposed hub uses three forwarding modes. **Connected multi-hop** is used when link quality and neighbor reachability are adequate. **Opportunistic replication** is used when multiple plausible contacts exist, with bounded copies and message expiry. **Store-carry-forward** is used when people or vehicles physically move messages across a communication gap. Every mode needs a hop limit, message identifier, deduplication cache, queue bound, priority policy, and congestion indicator.

Meshtastic demonstrates a practical low-cost pattern: inexpensive LoRa devices, local phone connections, rebroadcast, hop limits, and managed flooding 16. The hub can borrow those principles while making its safety policy explicit. NIST’s real-time mesh work also emphasizes rapid link-quality measurement and relay placement before a responder loses command connectivity 18.

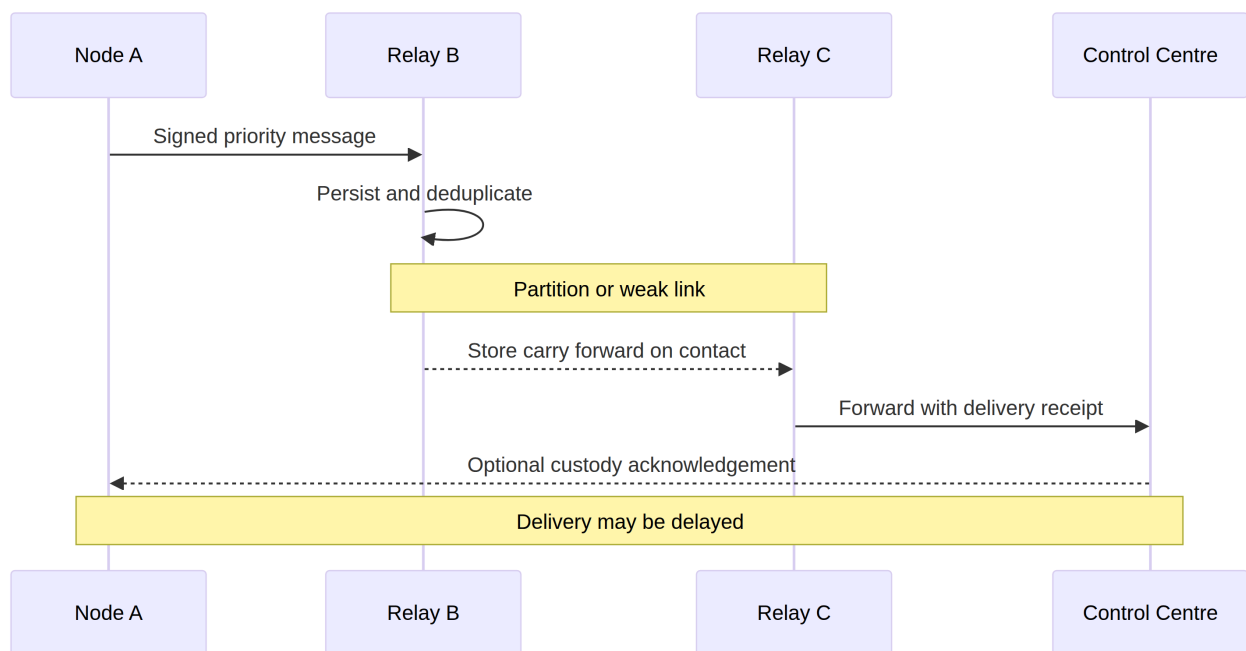


Figure 3. A message may travel immediately, wait in a queue, or move physically with a carrier before delivery.

The design must avoid the misleading claim that “mesh means guaranteed communication.” More relays can improve connectivity, but more transmitters also increase airtime and collisions. A useful target is graceful degradation: when nodes disappear, high-priority messages continue to receive bounded opportunities while routine traffic is delayed or rejected.

CHAPTER 3

PROPOSED SYSTEM AND DESIGN

3.1 Proposed System Architecture

The Dynamic Disaster Communication Hub is organized into five layers. The **physical and energy layer** contains the ESP32 controller, external LoRa radio, antenna, optional sensors, local storage, user interface, enclosure, and battery/UPS or solar input. The **link layer** provides raw LoRa, optional local Wi-Fi/BLE, and adapters for a local gateway or backhaul. The **message layer** provides durable records, identity, integrity, expiry, priority, provenance, and deduplication. The **routing layer** selects multi-hop, bounded flooding, opportunistic replication, or store-carry-forward. The **application layer** provides forms, dashboards, sensor checks, triage, and operator review.

A node can operate as a resident terminal, relay, sensor node, gateway, or command console depending on its role configuration. Role configuration should be authenticated and deny-by-default. A relay should not automatically gain authority to issue official warnings. A gateway should be treated as a trust boundary because it can translate, filter, or re-encrypt traffic.

The architecture is deliberately modular. The application should not depend on a particular radio driver. A later prototype may replace raw LoRa with a different link or add cellular and satellite backhaul without changing the message identity and priority model.

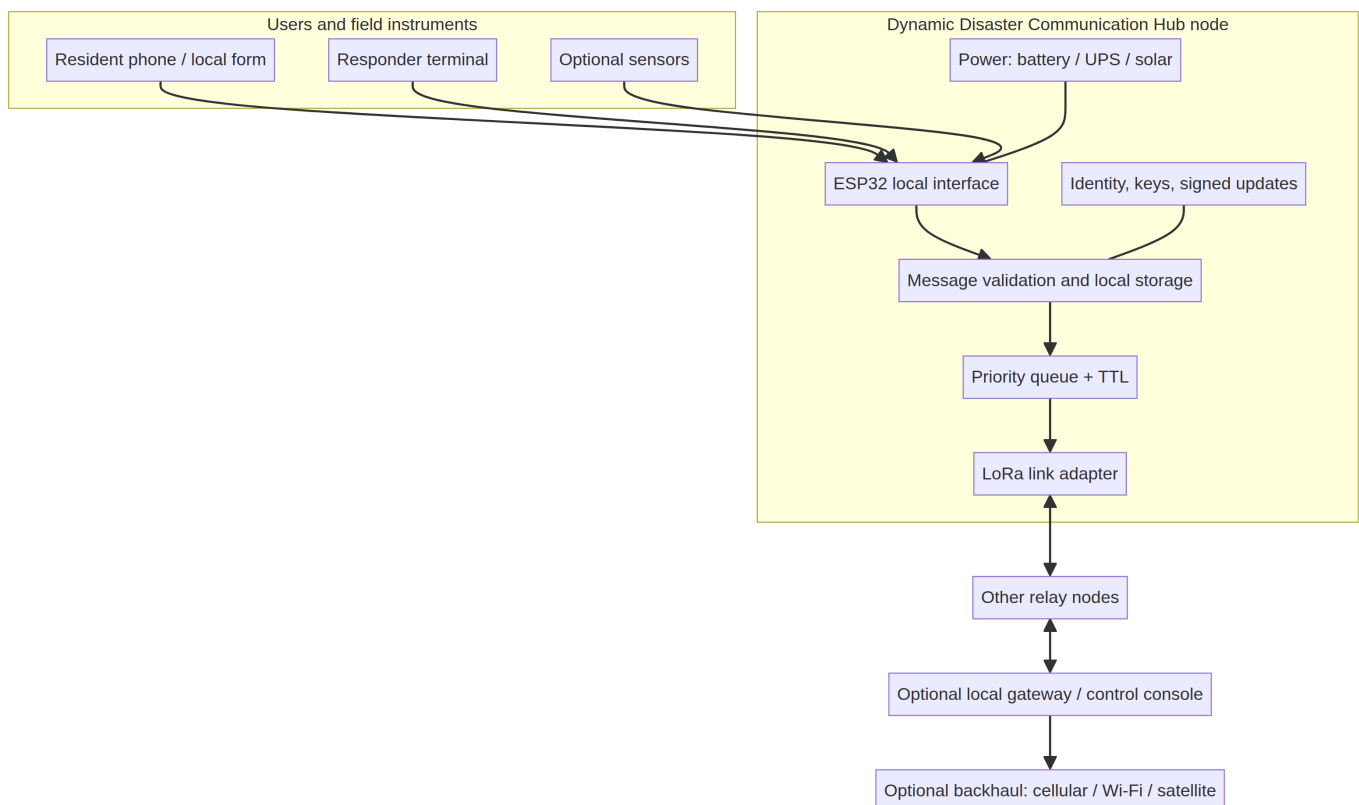


Figure 4. Proposed Dynamic Disaster Communication Hub architecture.

Component	Responsibility	Failure behavior
ESP32 controller	Local UI, sensor control, protocol execution	Continue queued operation or enter low-power mode
LoRa radio	Short authenticated field messages	Retry, queue, or use a different relay
Storage	Durable message and audit records	Preserve until expiry or capacity policy
Routing engine	Forwarding, deduplication, hop limits	Drop duplicates and low-priority traffic first
Power subsystem	Battery, charging, measurement	Announce low energy and reduce service
Gateway	Local aggregation and optional backhaul	Operate locally if backhaul fails
Operator console	Review, map, escalation, export	Use offline forms and local logs

3.2 Hardware and Energy Design

A minimum field node contains an ESP32 development board or custom board, an external LoRa transceiver, a regional antenna, a battery, a regulator, local non-volatile storage, and a protected enclosure. Optional elements include a small display, tactile buttons, a buzzer, environmental sensors, an accelerometer, a GPS receiver, or a secure element. The minimum viable prototype should start with two nodes and one relay before adding sensors or AI.

The antenna should be mounted so that the enclosure, user's body, wet ground, and nearby metal do not create an untested RF condition. Elevated relay placement is usually more valuable than claiming a larger theoretical range. Field tests should record packet delivery, received signal strength, signal-to-noise ratio, obstacles, antenna height, weather, and node orientation.

Power must be sized from measurements. Wi-Fi can be activated on demand for a local phone interface, then disabled. LoRa status beacons can be scheduled. Deep sleep is useful for event-driven nodes, but it disconnects Wi-Fi and Bluetooth [8](#). The power system should include burst-

current headroom, charging losses, cold-weather derating, retransmissions, display load, and gateway draw.

A rugged case should separate the RF antenna from the battery and provide a visible status indicator that does not require a network. A field service label should show node identifier, operating region, battery date, channel profile, emergency reset method, and support contact.



Figure 5. Conceptual field node visual; the technical block diagram should be drawn in the implementation record.

Suggested bill of materials: ESP32 board; external LoRa module; antenna and connector; battery and protection board; regulator; microSD or flash storage; enclosure; optional sensor kit; cable and spare antenna; and a printed quick-start card. Cost should be reported as a total cost of ownership rather than a radio-only bill of materials.

Software, data model, and message lifecycle

The software should treat every record as a durable object. A proposed record contains:

`message_id`, `sender_id`, `sender_role`, `created_at`, `expires_at`, `priority`, `type`, `location`, `location_uncertainty`, `payload`, `provenance`, `hop_limit`, `replication_budget`, `integrity_tag`, and `schema_version`. A payload may be a compact structured form or a short text field. The location field should support “unknown” and uncertainty rather than forcing a false precision.

The lifecycle is: compose or ingest; validate schema; attach time, role, and provenance; authenticate; persist to non-volatile storage; schedule transmission; forward; deduplicate; acknowledge selectively; deliver to an operator or peer; escalate if required; expire; and audit. A node should persist a message before transmitting it so a power interruption does not erase the only copy.

Acknowledgements must be used carefully. A receipt can confirm that a neighbor stored a message, not that a rescue action occurred. The interface should distinguish **received**, **forwarded**, **seen by operator**, and **action confirmed**. Confusing these states can create dangerous reassurance.

The queue should reserve capacity for life-safety and medical messages. Routine administrative traffic can be delayed or rejected during a burst. A time-to-live prevents stale evacuation or route information from circulating indefinitely. A deduplication cache prevents managed flooding from multiplying identical records.

Proposed message classes:

Class	Examples	Default policy
P0 life safety	trapped person, immediate hazard	Immediate attempt, high replication, operator alert
P1 medical / evacuation	urgent medical need, route closure	High priority, bounded retries
P2 logistics	water, shelter capacity, fuel	Normal priority, store-and-forward
P3 sensor status	battery, temperature, link health	Periodic, suppress duplicates
P4 administration	configuration, routine notes	Delay or reject under congestion

3.4 Routing and Congestion Control

The routing engine should choose a delivery policy from the available evidence. Neighbor discovery can record received signal strength, signal-to-noise ratio, recent delivery history, battery state, role, and queue occupancy. A relay with a good link but nearly empty battery should not be treated as an unlimited router.

A practical first implementation is bounded managed flooding. Each packet carries a hop limit and a message identifier. A node forwards only if it has not seen the identifier, the message has

not expired, the priority is admitted, and the node has queue capacity. A randomized contention window reduces simultaneous rebroadcast. A node that hears a stronger or farther relay forwarding the same packet can suppress its own duplicate. This approach is simple but must be tested for broadcast storms.

A later implementation can add route scoring. The score may combine recent delivery probability, link quality, estimated delay, battery, role, and mission priority. The score must remain explainable because responders need to know why a route was selected.

Traffic control is as important as topology. LoRa capacity is limited by data rate, airtime, spreading factor, collision probability, and message length 9–12. The system should fragment only when necessary, limit routine telemetry, suppress repeated sensor values, and apply admission control during an emergency burst. It should surface queue age and dropped-message counts to operators.

Control	Purpose	Example rule
Hop limit	Bound network-wide work	P0: 6; P3: 2, subject to test
TTL	Prevent stale information	Route and evacuation records expire
Deduplication	Stop duplicate storms	Cache IDs for message lifetime
Priority queue	Protect life-safety traffic	Reserve buffer for P0/P1
Rate limit	Protect airtime	Cap routine reports per node
Replication budget	Limit copies	P0 bounded copies; P4 one copy

AI and IoT decision-support layer

AI is valuable when it reduces delay or organizes information without hiding uncertainty. The proposed hub uses AI in bounded roles: sensor-health checks, threshold and anomaly screening, feature extraction, message classification, probabilistic victim-search areas, and operator prioritization. It does not autonomously order evacuation, dispatch a rescue team, authorize entry into unstable structures, or replace an official warning authority.

IoT sensing should preserve provenance. Each reading should carry sensor identity, timestamp, calibration status, location uncertainty, missingness, and model version if an AI rule is applied. A disagreement between sensors should reduce confidence or trigger review rather than be

silently averaged. A deterministic fallback must remain available when the model is unavailable, the network is partitioned, or the sensor health is uncertain.

Research supports sensor fusion, early-warning systems, edge processing, crisis-message classification, and probabilistic localization, but it also documents false alarms, latency, coverage gaps, non-line-of-sight limitations, and model uncertainty 19–24. WMO and NIST guidance emphasizes human expertise, representative testing, monitoring, and safe degradation 25.

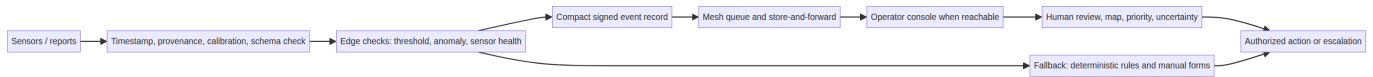


Figure 7. Proposed AI workflow with fallback and human review.

The operator console should show the original report, the model output, evidence, confidence, recency, and disagreement. A high-recall triage model may send more false positives to humans because missing a genuine help request can be worse than reviewing extra candidates. Every model output and human action should be logged.

AI acceptance principles: explainability over novelty; high-recall screening for distress; no silent deletion of low-confidence reports; model versioning; drift monitoring; multilingual and accessibility testing; an override and shutdown path; and a named human owner for every life-safety action.

3.6 Security, Privacy, and Interoperability

A disaster node is safety-sensitive IoT. It can affect the physical world, may be physically captured, and may be unable to reach a central security service. NIST identifies device, data, and privacy risks for IoT systems 28. Its device capability catalog recommends unique identification, authorized configuration, data protection, restricted interfaces, secure updates, security-state awareness, and integrity protection 29.

The raw-LoRa mesh should authenticate application messages and protect against replay. A device identity should be unique and revocable. Sensitive location should be minimized, retained only as long as necessary, and disclosed according to role. The operator console should record access, exports, and configuration changes. A captured relay should not expose every historical message or a master key.

When LoRaWAN is used, its network and application keys provide a useful reference model, but secure implementation still requires proper key generation, storage, and nonce handling 30. When traffic crosses a gateway into MCX, LMR, cellular, or satellite systems, the translation point becomes a trust boundary. 3GPP documents how interworking can terminate one

encryption context and apply another 33. The deployment record must state where encryption ends, how identity maps, and what priority semantics survive translation.

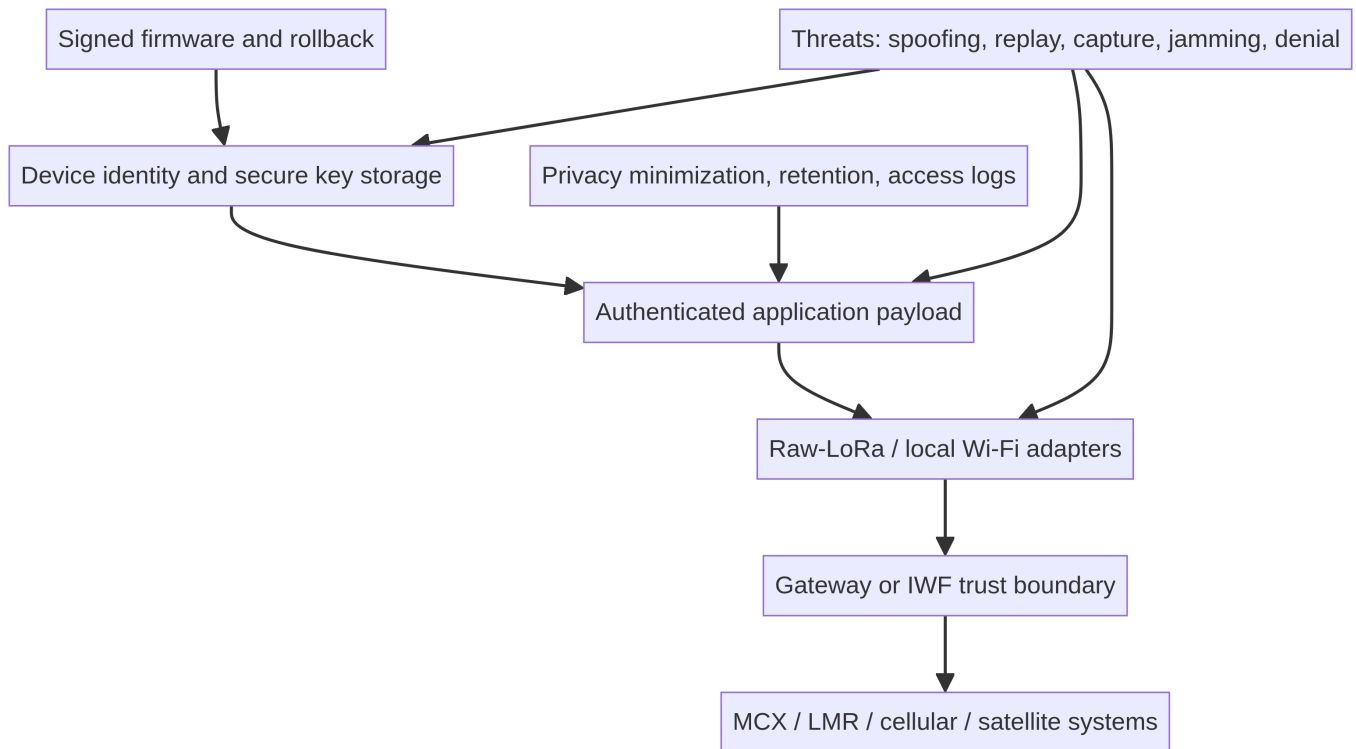


Figure 8. Trust boundaries and controls.

Threat	Control	Residual limitation
Spoofing	Device identity and signed records	Compromised credentials remain possible
Replay	Nonce, timestamp, expiry, deduplication	Clock uncertainty must be managed
Eavesdropping	Encryption and minimal data	Metadata may remain visible
Node capture	Secure storage, revocation, least privilege	Physical access cannot be eliminated
Jamming / interference	Channel plan, fallback, logging	Radio silence may remain unavoidable
Malicious update	Signed firmware and rollback	Key compromise requires recovery

CHAPTER 4

IMPLEMENTATION AND EVALUATION

4.1 Operating Workflow and Field Deployment

Preparedness begins before the hazard. Authorities or campus teams should inventory nodes, register identities, load regional channel profiles, test batteries, print offline procedures, designate relay locations, and exercise the message forms. The plan should identify who can issue official warnings, who can review reports, how vulnerable groups are included, and how the hub connects to authorized emergency systems.

During activation, nodes perform self-tests for battery, storage, clock, sensors, radio, and security state. They discover neighbors and estimate link quality. Users submit structured records. The node validates and persists each record before transmission. Relays deduplicate and forward according to priority, hop limit, expiry, battery, and queue rules. When a gateway or responder reaches a control point, queued records synchronize. Operators review uncertainty and issue actions through authorized procedures.

Recovery includes credential rotation, equipment inspection, log export, battery replacement, damage assessment, and after-action review. A system that works only during a demonstration but cannot be maintained, reset, and audited is not operationally resilient.

Field rules: elevate relay antennas where safe; avoid placing nodes in flood paths; label every node; record the operating region and channel plan; keep a paper fallback; do not publish private location data broadly; do not interpret a message receipt as a completed rescue; and do not enter a hazardous zone solely because an AI score is high.

Graceful-degradation ladder:

1. Backhaul connected: synchronize with official systems.
2. Local gateway only: retain local dashboard and mesh queues.
3. Mesh partitioned: use multi-hop and store-and-forward.
4. Sparse contacts: use responder-carried messages.
5. Power constrained: protect P0/P1 traffic and reduce telemetry.
6. Radio unavailable: revert to paper, voice, satellite, LMR, or other authorized channels.

4.2 Implementation Plan and Prototype

A staged prototype reduces risk. **Stage 1** assembles two ESP32-plus-LoRa nodes and verifies point-to-point messages, device identity, storage, and power measurement. **Stage 2** adds a relay and implements message IDs, hop limits, deduplication, TTL, priority queues, and random contention. **Stage 3** adds a local Wi-Fi or Bluetooth interface for structured forms while

preserving LoRa as the field link. **Stage 4** adds sensor simulation and deterministic anomaly rules. **Stage 5** adds an operator console, optional AI triage, and gateway synchronization.

The minimum viable demonstration should show: a resident distress record; a responder status record; a relay that stores and forwards a message after a temporary partition; duplicate suppression; a queue that protects high-priority traffic; and recovery after a node reboot. The demonstration should also show a failure case: a depleted relay battery, a missing gateway, or a burst of routine telemetry.

Configuration must be reproducible. The project record should include firmware version, radio module, antenna, regional frequency profile, transmit power, bandwidth, spreading factor, coding rate, payload size, node roles, queue limits, hop limits, TTLs, and test locations. A “range” claim without these parameters is not meaningful.

The project team should keep hardware and software interfaces modular. A radio adapter should expose send, receive, link estimate, and power-state functions. The message layer should not know whether a record moved through raw LoRa, local Wi-Fi, or a future gateway. The operator interface should show the message’s delivery state and uncertainty without inventing certainty.

The implementation should be reviewed by an advisor familiar with radio regulation, electrical safety, and disaster-management practice before field use. The prototype is suitable for a campus or controlled field trial, not unsupervised deployment in a live emergency.

4.3 Evaluation Methodology

Evaluation should compare the hub with a single-hop baseline and define the operating envelope. The envelope includes geography, obstacles, node count, relay positions, payload size, traffic burst, channel settings, battery state, and weather. Repeated trials are needed because radio performance is variable.

Core metrics: Data Extraction Rate (DER) or Packet Delivery Ratio (PDR); end-to-end and queueing latency; duplicate and drop rate; goodput; coverage and link margin; energy per successfully delivered message; battery endurance; simultaneous-node capacity; gateway-loss recovery time; and cost per covered area or delivered message. A DER target above 0.95 can be meaningful only after geography and traffic are defined [35](#).

Metric	Measurement	Test conditions	Qualified acceptance example
DER / PDR	Delivered unique records ÷ sent records	Fixed route and burst levels	Target declared per envelope
Latency	Creation to operator receipt	P0 and P2 traffic separately	Report median and tail
Duplicate rate	Duplicate forwards ÷ forwards	Managed flooding burst	Remain below set airtime budget
Energy/message	Joules ÷ successful records	Battery and retransmission states	Compare modes
Coverage	Location and link margin map	Open, urban, obstructed	Publish tested area only
Recovery time	Time to resume after relay loss	Remove gateway or node	Show queue behavior
Capacity	Successful records under load	Increasing concurrent nodes	Identify graceful limit

Test scenarios should include normal traffic, emergency bursts, a partition, gateway loss, low battery, reboot, packet corruption, interference simulation, and stale-message expiry. AI tests should include sensor drift, missing inputs, contradictory reports, rare-event conditions, and operator override. A result table must distinguish measured values from design assumptions.

4.4 Results Presentation and Risk Analysis Plan

The final prototype report should present results as distributions or repeated-trial summaries rather than a single best-case distance. Coverage maps should show tested points and failure points. Queue traces should show how priority classes behave during a burst. Energy profiles should identify sleep, receive, transmit, display, storage, and gateway components. A failure-injection table should record what happened when a relay disappeared, a battery fell below threshold, or a duplicate storm was triggered.

The principal risk is not that the hub fails in an obvious way. It is that it appears to work while silently dropping or delaying the information that matters most. The test plan must therefore

measure missing messages, stale messages, incorrect confidence, identity failures, and operator misunderstandings.

Failure mode	Detection signal	Mitigation	Human safeguard
Dead battery	Voltage and heartbeat absent	Power warning, protect P0/P1	Field inspection
Antenna damage	Falling link margin	Spare antenna, relay relocation	Do not assume coverage
Duplicate storm	Queue and airtime rise	Hop limits, suppression, rate limits	Reduce routine traffic
Gateway loss	Backhaul heartbeat absent	Local dashboard and DTN queue	Use alternate channel
Misleading sensor	Cross-sensor disagreement	Health checks, calibration	Require verification
Model drift	Confidence and error shift	Versioning and monitoring	Disable model if unsafe
Node capture	Identity anomaly	Revocation and rotation	Treat reports cautiously

CHAPTER 5

DISCUSSION AND CONCLUSION

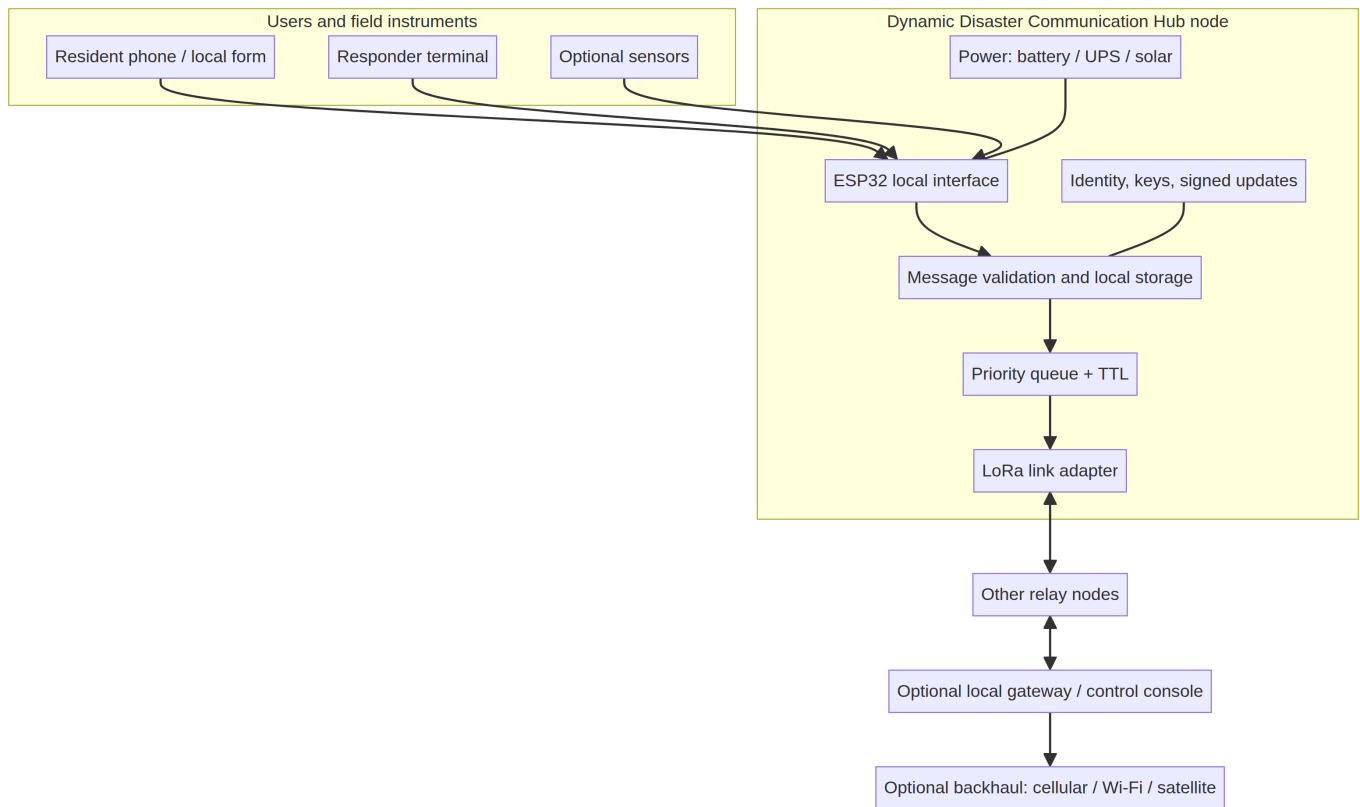


Figure 10. The evaluation dashboard should connect delivery, queue, energy, and failure state; the architecture visual above is a placeholder for the final dashboard layout.

The report should explicitly state what was not tested. A campus trial cannot prove performance in rubble, deep floodwater, cyclone winds, dense forest, or a city-wide outage. Honest limitations increase the value of the engineering result.

5.1 Discussion and Design Alternatives

The proposed hub improves resilience in four bounded ways. It creates a local communication path that does not require a functioning cellular tower. It allows messages to survive temporary disconnection through durable queues. It supports low-power status exchange over a small, replaceable node. It makes degradation visible through priority, queue, expiry, and link-quality information.

The hub does not restore damaged fiber, provide high-bandwidth media, replace public warning authorities, guarantee victim localization, or eliminate the need for trained responders. Its value depends on node density, antenna placement, battery condition, physical access, protocol configuration, and community adoption. A sparse mesh with no trained operator may be less useful than a well-maintained satellite terminal or LMR radio.

Design alternatives should be compared rather than treated as competitors. A cellular-fallback solution is stronger when towers and backhaul survive but fails under the same physical and power conditions the project addresses. A local LoRaWAN installation gives managed operations but requires gateway and server components. Raw LoRa mesh provides infrastructure independence but requires custom security and congestion control. A responder-carried DTN can cross large gaps but introduces delay and depends on movement. A hybrid system is therefore preferable: local raw-LoRa messaging, optional local gateway, and multiple external backhauls when available.

AI should remain subordinate to communications and operations. In a disaster, a simple, explainable message queue may be more valuable than a sophisticated model that cannot receive data or cannot explain its confidence. The strongest architecture therefore uses deterministic forms and rules as the base, then adds AI where it improves triage without becoming a single point of failure.

Whole-community design matters. Vulnerable groups may have limited power, language access, disability access, or device ownership. The hub should support trained intermediaries, audible and visual cues, multilingual forms, and paper procedures.

5.2 Conclusion and Future Work

This report proposed a Dynamic Disaster Communication Hub for conditions in which mobile towers and internet services are unavailable or intermittent. The design combines an ESP32 controller, an external LoRa radio, durable local storage, battery or solar power, controlled multi-hop forwarding, delay-tolerant store-and-forward delivery, and optional AI-assisted IoT decision support.

The central conclusion is qualified: **a layered ESP32-plus-LoRa system is a plausible supplementary communication and sensing aid when it is designed for short messages, bounded airtime, redundancy, security, human oversight, and measured failure behavior.** It should be presented as a local resilience layer, not as a universal substitute for public-safety communications.

Future work should: (1) validate the design under regional radio rules; (2) measure delivery and energy with complete assembled nodes; (3) compare managed flooding and route scoring; (4) test relay placement in urban, rural, and obstructed terrain; (5) add secure hardware and credential recovery; (6) evaluate multilingual and accessible forms; (7) test sensor fusion and model fallback; (8) conduct authorized interoperability exercises with cellular, LMR, MCX, satellite, and emergency-management systems; and (9) publish after-action findings with limitations.

A responsible prototype has clear stop conditions. If a model becomes unreliable, it can be disabled. If a node is captured, its credentials can be revoked. If airtime becomes congested, routine traffic can be rejected. If the mesh fails, operators can use paper, voice, satellite, LMR, or other authorized channels. This ability to fail safely is the defining engineering requirement.

Final design principle: The hub should make useful information more likely to arrive, more clearly labeled, and more safely handled when ordinary infrastructure fails. It should never create false confidence that communication, rescue, or warning delivery is guaranteed.

REFERENCES